

of workers who lose benefits when the PBGC takes over their plans, the average benefit reduction is 28%.³¹

In return for this guarantee, pension plans pay premiums to the PBGC, which in 2012 were \$9 per member for multiemployer plans and \$35 per member for fully funded single-employer plans. Underfunded plans pay more. As is so often the case when governments provide insurance, PBGC premiums are way below the true cost of the coverage. The Congressional Budget Office estimates that premiums should be increased by more than six times to cover the shortfall from projected future claims, and this does not even include underfunding on the existing claims covered by the PBGC.³²

PBGC protection is a *put option* for employers: when things are going really badly, you let the government pick up the tab.³³ The PBGC floor leads to an *incentive problem*: the pension fund manager can take *excessive risk* and the firm has an incentive to *underfund* because taxpayers ultimately bear the pension risk. If the bets work out, that's wonderful—the company gets a *contribution holiday* and in some cases can even extract money from the pension plan.³⁴ If things turn out disastrously, you let the PBGC mop up the mess. Many companies, like United Airlines and American Airlines, entered bankruptcy partly to shed their pension liabilities. These didn't disappear—the taxpayer picked up the tab.

3.3. PENSION UNDERFUNDING

The big drop in asset prices during 2008–2009 was not kind to pension funds. Milliman, a pension consulting firm, tracks the one hundred largest defined benefit plans sponsored by American public companies. Figure 1.3 plots the funding status of the Milliman 100 from 1999–2011.³⁵ The health of these plans has suffered twice in the last decade, mostly due to steep declines in asset values (see also Figure 1.2). In fiscal year 2011, the Milliman 100 companies had a record funding deficit of \$327 billion, corresponding to a funding ratio of 79%. The funding ratios in 2007 were positive, at 105%, and despite equity markets recovering post-2009, pension funds did not fully bounce back—partly because pension liabilities grew faster than recovering asset values.

³¹ See the 2008 report, PBGC's Guarantee Limits—An Update, published by the PBGC.

³² See Congressional Budget Office, The Risk Exposure of the Pension Benefit Guaranty Corporation, 2005. For a summary of academic work, see Brown (2008).

³³ This is first noted by Sharpe (1976) and Treynor (1977). Sharpe and Treynor play large roles in developing the CAPM, the first model of factor risk (see chapter 6). Pennacchi and Lewis (1994) compute explicit values for the PBGC put option guaranty.

³⁴ Section 4980 of the Internal Revenue Code makes it extremely costly for a company to simply close an overfunded pension plan and collect the value of the pension fund in excess of its liabilities (or surplus) by imposing an excise tax of 50%.

³⁵ Data from the Milliman 2012 Pension Funding Study.